

1. Title

ShipTrack Pro : A Full-Stack Shipment Tracking & Delivery Visibility Platform that enables businesses, logistics providers, and customers to track shipments in real time, monitor delivery progress, manage proof of delivery, receive notifications, and analyze logistics performance through a centralized platform.

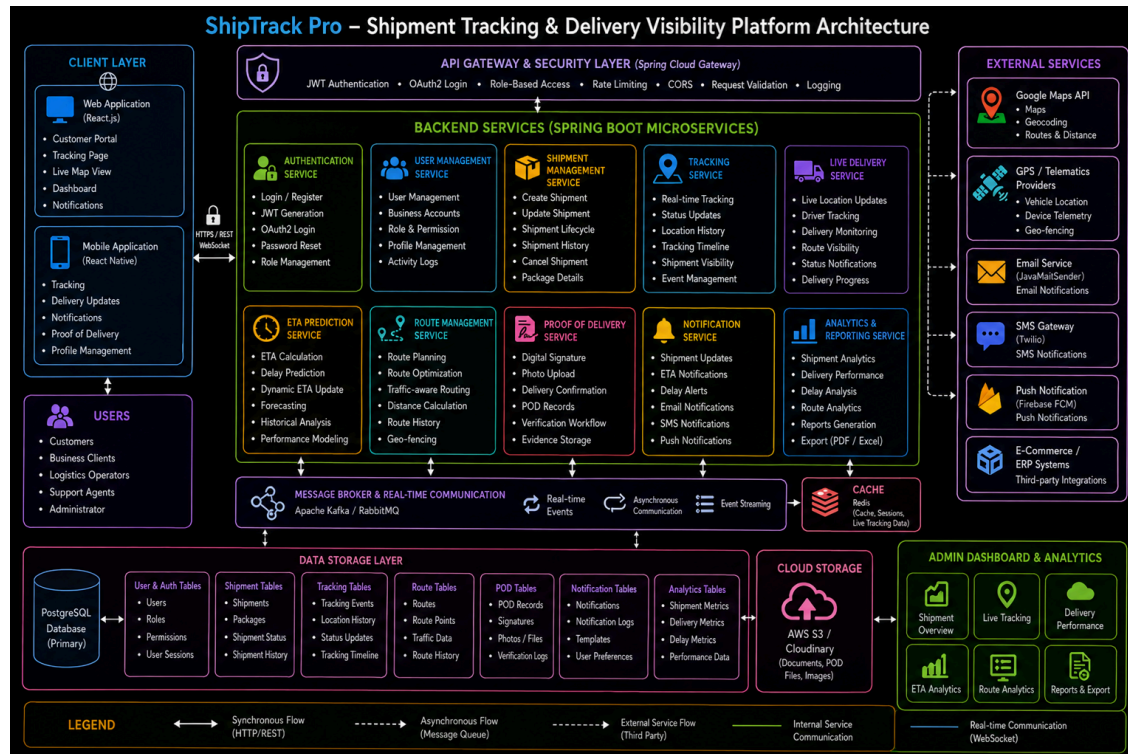
2. Objective

Build a full-stack web application (React.js frontend + Spring Boot backend) that allows logistics companies and customers to monitor shipments throughout the delivery lifecycle. The platform provides real-time shipment tracking, ETA estimation, proof of delivery management, route monitoring, delivery notifications, and logistics analytics through an integrated dashboard.

Outcomes

- (i) Developed and deployed a full-stack React + Spring Boot application.
- (ii) Implemented secure user authentication using JWT and OAuth2.
- (iii) Built shipment creation and tracking functionality.
- (iv) Implemented real-time delivery monitoring and route tracking.
- (v) Added ETA prediction and delay monitoring systems.
- (vi) Created proof of delivery verification workflows.
- (vii) Implemented analytics dashboards and reporting modules.
- (viii) Deployed frontend and backend using PostgreSQL in production.
- (ix) Added notification systems for shipment updates and delivery alerts.

3. Architecture Components



4. Modules to be Implemented

1. User Authentication & Role-Based Access

- (i) JWT Authentication
- (ii) OAuth2 Login
- (iii) Password Reset
- (iv) Profile Management
- (v) Role Management

Roles

- Customer
- Business Client
- Logistics Operator
- Support Agent
- Administrator

2. User Management Module

- (i) Customer registration
- (ii) Business account creation
- (iii) Profile management
- (iv) Account settings
- (v) User activity tracking
- (vi) Role-based access control

3. Shipment Management Module

- (i) Create shipment
- (ii) Update shipment details
- (iii) Shipment lifecycle management
- (iv) Shipment history
- (v) Package information management
- (vi) Shipment cancellation

Shipment Information

- Tracking Number
- Sender Details
- Receiver Details
- Package Details
- Shipment Status
- Delivery Address

4. Shipment Tracking Module

- (i) Real-time tracking
- (ii) Shipment status updates

- (iii) Tracking history
- (iv) Location monitoring
- (v) Tracking timeline
- (vi) Shipment visibility dashboard

Tracking Status

- Created
- Picked Up
- In Transit
- Out for Delivery
- Delivered
- Failed Delivery

5. Live Delivery Monitoring Module

- (i) Live location updates
- (ii) Delivery monitoring dashboard
- (iii) Driver tracking
- (iv) Route visibility
- (v) Delivery status notifications
- (vi) Delivery progress monitoring

6. ETA Prediction Module

- (i) Delivery time estimation
- (ii) Delay prediction
- (iii) Dynamic ETA updates
- (iv) Historical delivery analysis
- (v) Delivery forecasting

(vi) Route performance analysis

7. Route Management Module

(i) Route planning

(ii) Route optimization

(iii) Route history

(iv) Distance calculations

(v) Traffic-aware routing

(vi) Route analytics

8. Proof of Delivery Module

(i) Digital signature capture

(ii) Delivery photo upload

(iii) Delivery confirmation

(iv) Verification workflows

(v) POD record management

(vi) Delivery evidence storage

9. Notification Module

(i) Shipment updates

(ii) ETA notifications

(iii) Delivery alerts

(iv) Delay warnings

(v) Email notifications

(vi) SMS notifications

(vii) Push notifications

10. Analytics Dashboard Module

Customer Dashboard

- (i) Active shipments
- (ii) Shipment history
- (iii) Delivery status overview
- (iv) Notification center
- (v) Tracking insights

Business Dashboard

- (i) Shipment analytics
- (ii) Delivery performance
- (iii) Delay analysis
- (iv) Logistics overview
- (v) Customer activity

Admin Dashboard

- (i) User management
- (ii) Shipment monitoring
- (iii) Delivery analytics
- (iv) Route performance
- (v) System monitoring
- (vi) Reports management

11. Reports & Export Module

- (i) Shipment reports
- (ii) Delivery reports
- (iii) Route performance reports
- (iv) Delay analysis reports
- (v) Logistics reports
- (vi) PDF export
- (vii) Excel export

12. Final Integration, Testing & Deployment

5. Week-wise Module Implementation and High-Level Requirements

Milestone 1: Week 1 & 2 — Project Initialization, Design Process & Core Setup

Tasks

- (i) Define logistics workflows and project objectives.
- (ii) Design system architecture and database schema.
- (iii) Create UI wireframes and workflow planning.
- (iv) Setup React frontend and Spring Boot backend.
- (v) Implement JWT authentication and role-based access control.
- (vi) Build shipment management workflows.
- (vii) Develop shipment tracking dashboard.

Outcomes

- (i) Understand shipment tracking and logistics workflows.
- (ii) Learn system architecture and database design concepts.
- (iii) Build frontend and backend project initialization.
- (iv) Working authentication and shipment tracking system.

Milestone 2: Week 3 & 4 — Live Delivery Monitoring & ETA Prediction

Tasks

- (i) Integrate Google Maps API.
- (ii) Build live delivery tracking workflows.
- (iii) Develop shipment status management.
- (iv) Implement ETA calculation logic.
- (v) Generate delivery forecasts.
- (vi) Build delay prediction workflows.

Outcomes

- (i) Implement live delivery monitoring systems.
- (ii) Build ETA prediction and forecasting workflows.
- (iii) Understand logistics visibility and route tracking concepts.
- (iv) Generate real-time delivery monitoring insights.

Milestone 3: Week 5 & 6 — Proof of Delivery & Analytics

Tasks

- (i) Build digital delivery confirmation workflows.
- (ii) Implement signature capture and verification.

- (iii) Develop proof of delivery system.
- (iv) Build shipment analytics dashboards.
- (v) Generate delivery performance reports.
- (vi) Implement route history tracking.

Outcomes

- (i) Build proof of delivery and verification systems.
- (ii) Implement logistics analytics and reporting workflows.
- (iii) Understand delivery performance monitoring concepts.
- (iv) Complete end-to-end shipment visibility workflows.

Milestone 4: Week 7 & 8 — Testing, Deployment & Documentation

Tasks

- (i) Perform application testing and workflow validation.
- (ii) Improve tracking performance and UI responsiveness.
- (iii) Deploy platform using Docker and cloud environments.
- (iv) Prepare final project documentation and presentation.
- (v) Demonstrate the complete ShipTrack Pro platform.

Outcomes

- (i) Gain deployment and testing experience.
- (ii) Improve platform stability and usability.
- (iii) Complete live deployment and final demonstration.
- (iv) Prepare professional project documentation and presentation.

6. Evaluation Criteria

Milestone 1 (Week 2)

- (i) Project initialization and architecture setup completed.
- (ii) Authentication and shipment tracking workflows implemented.
- (iii) Shipment management dashboard functional.
- (iv) System design and UI planning completed.

Milestone 2 (Week 4)

- (i) Live delivery monitoring workflows implemented.
- (ii) ETA prediction and delay forecasting system functional.
- (iii) Maps and location services integration working.
- (iv) Delivery status tracking integrated.

Milestone 3 (Week 6)

- (i) Proof of delivery system implemented.
- (ii) Delivery verification workflows functional.
- (iii) Analytics dashboard and performance reports generated.
- (iv) Route history tracking integrated.

Milestone 4 (Week 8)

- (i) Fully deployed frontend and backend.
- (ii) Testing and validation completed.
- (iii) Documentation and presentation prepared.
- (iv) Successful end-to-end platform demonstration completed.

7. Tools & Tech Stack

Programming Language

Backend

- (i) Java
- (ii) Spring Boot

Frontend

- (i) JavaScript
- (ii) React.js

Database

Primary Database

- (i) PostgreSQL

Caching

- (i) Redis

Authentication

- (i) Spring Security
- (ii) JWT Authentication
- (iii) OAuth2 Login

Maps & Tracking

- (i) Google Maps API
- (ii) OpenStreetMap API

Libraries & Frameworks

Backend

- (i) Spring Boot
- (ii) Spring Security
- (iii) Spring Data JPA
- (iv) Hibernate
- (v) Maven
- (vi) Spring WebSocket

Frontend

- (i) React.js
- (ii) React Router
- (iii) Axios
- (iv) Tailwind CSS
- (v) Chart.js
- (vi) Context API

Notifications

- (i) Firebase Cloud Messaging
- (ii) JavaMailSender
- (iii) Twilio SMS

Storage

- (i) AWS S3
- (ii) Cloudinary

Testing

- (i) JUnit
- (ii) Mockito
- (iii) Postman
- (iv) React Testing Library

Dev & Deployment Tools

- (i) IntelliJ IDEA
- (ii) VS Code
- (iii) Git & GitHub
- (iv) Docker
- (v) Docker Compose
- (vi) GitHub Actions
- (vii) AWS / Azure
- (viii) Nginx
- (ix) Postman